

```
1 !pip install -q yfinance
```

```
1 import yfinance as yf
2 import pandas as pd
3 df = yf.download('BTC-USD', period='5y')
4 display(df.head())
```

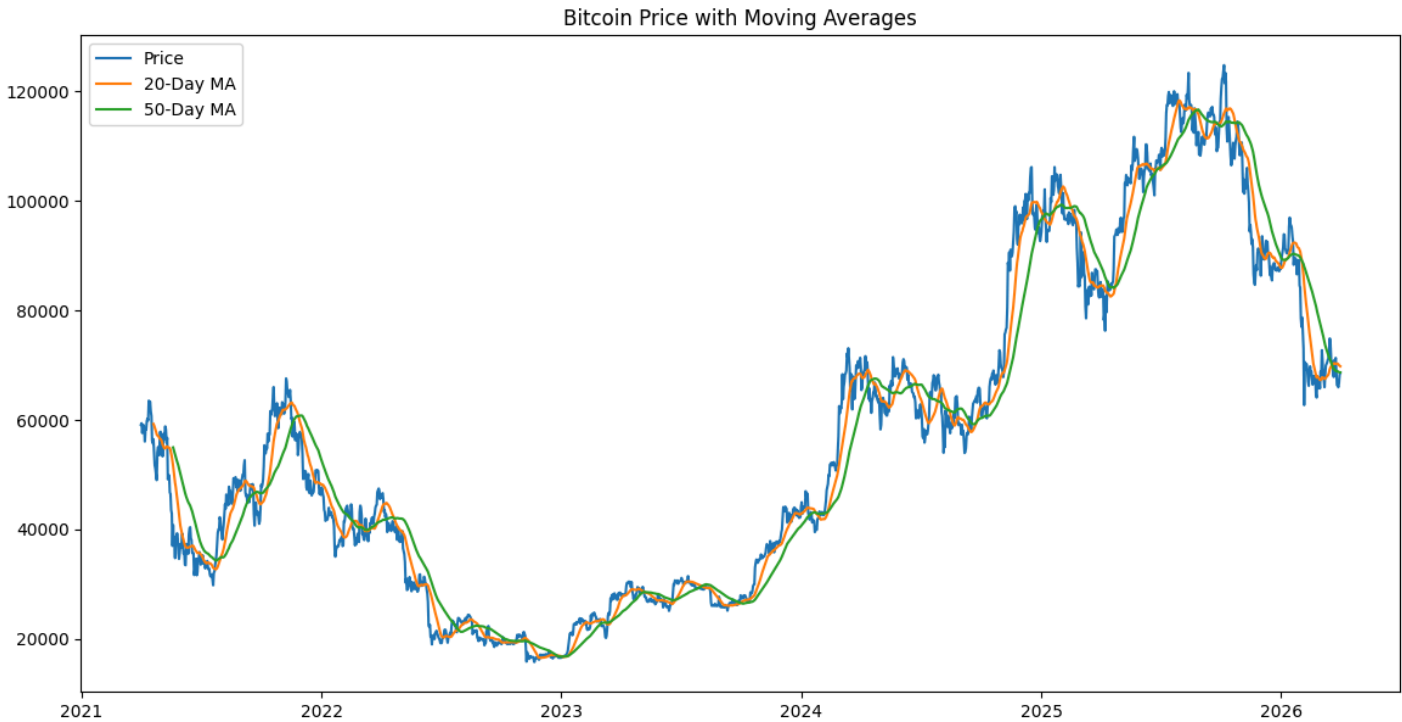
/tmp/ipykernel_1711/1761585211.py:3: FutureWarning: YF.download() has changed argument auto_adjust default to True
df = yf.download('BTC-USD', period='5y')
[*****100%*****] 1 of 1 completed

Price	Close	High	Low	Open	Volume	
Ticker	BTC-USD	BTC-USD	BTC-USD	BTC-USD	BTC-USD	
Date						
2021-04-01	59095.808594	59586.070312	58505.277344	58926.562500	61669163792	
2021-04-02	59384.312500	60267.187500	58869.281250	59098.878906	58727860620	
2021-04-03	57603.890625	60110.269531	57603.890625	59397.410156	59641344484	
2021-04-04	58758.554688	58913.746094	57168.675781	57604.839844	50749662970	
2021-04-05	59057.878906	59891.296875	57694.824219	58760.875000	60706272115	

```
1 df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
DatetimeIndex: 1827 entries, 2021-04-01 to 2026-04-01
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype
---  -
0   (Close, BTC-USD) 1827 non-null   float64
1   (High, BTC-USD)  1827 non-null   float64
2   (Low, BTC-USD)   1827 non-null   float64
3   (Open, BTC-USD)  1827 non-null   float64
4   (Volume, BTC-USD) 1827 non-null   int64
dtypes: float64(4), int64(1)
memory usage: 85.6 KB
```

```
1 df['MA20'] = df['Close'].rolling(window=20).mean()
2 df['MA50'] = df['Close'].rolling(window=50).mean()
3 plt.figure(figsize=(14, 7))
4 plt.plot(df['Close'], label='Price')
5 plt.plot(df['MA20'], label='20-Day MA')
6 plt.plot(df['MA50'], label='50-Day MA')
7 plt.title('Bitcoin Price with Moving Averages')
8 plt.legend()
9 plt.show()
```



```
1 import matplotlib.pyplot as plt
2 plt.figure(figsize=(14, 5))
3 plt.bar(df.index, df['Volume'].iloc[:, 0], color='orange')
4 plt.title('Bitcoin Trading Volume')
5 plt.grid(axis='y')
6 plt.show()
```

TypeError Traceback (most recent call last)

[/tmp/ipykernel_1711/2569975663.py](#) in <cell line: 0>()

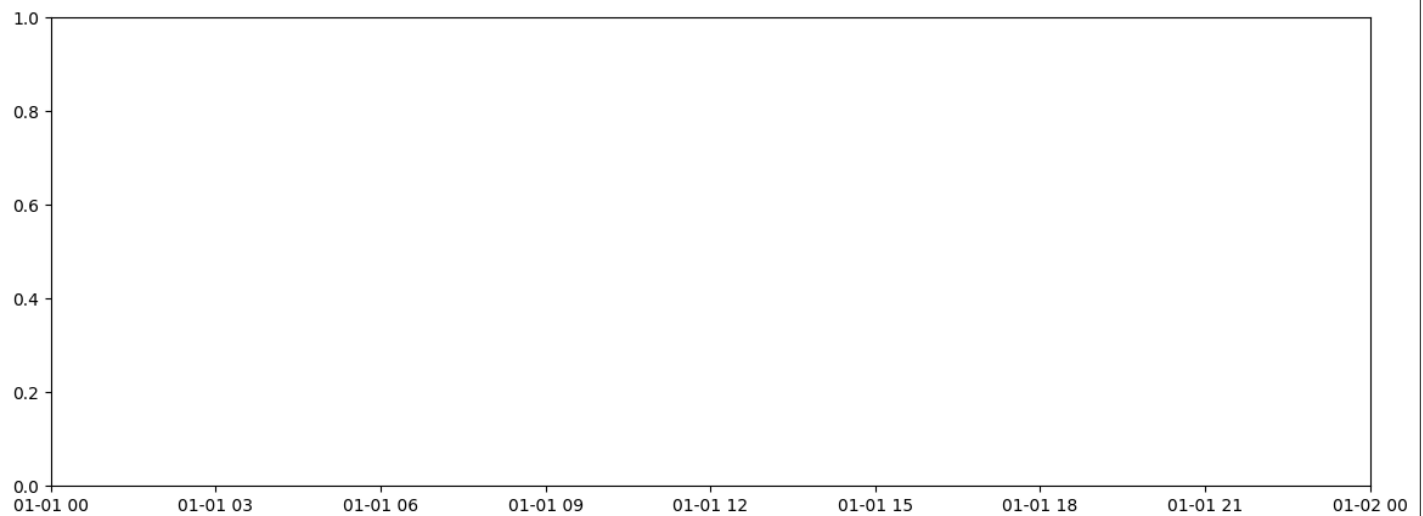
```
1 plt.figure(figsize=(14, 5))
----> 2 plt.bar(df.index, df['Volume'], color='orange')
      3 plt.title('Bitcoin Trading Volume')
      4 plt.grid(axis='y')
      5 plt.show()
```

⬆ 5 frames

[/usr/local/lib/python3.12/dist-packages/matplotlib/patches.py](#) in set_linewidth(self, w)

```
437     if w is None:
438         w = mpl.rcParams['patch.linewidth']
--> 439     self._linewidth = float(w)
440     self._dash_pattern = mlines._scale_dashes(
441         *self._unscaled_dash_pattern, w)
```

TypeError: only length-1 arrays can be converted to Python scalars



Next steps: [Explain error](#)

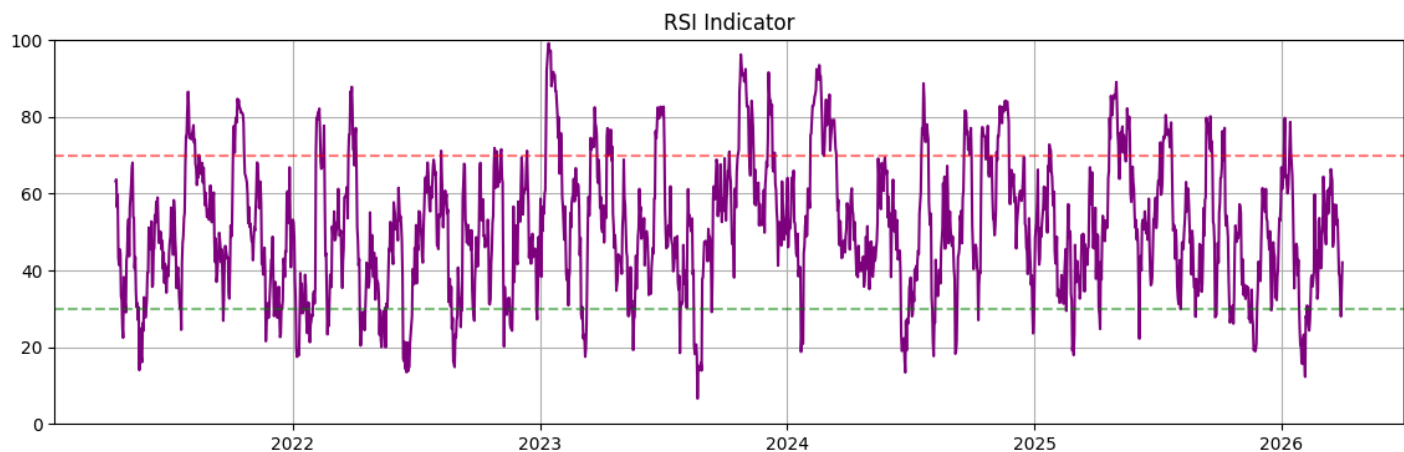
```
1 df['MA20'] = df['Close'].iloc[:, 0].rolling(window=20).mean()
2 df['MA50'] = df['Close'].iloc[:, 0].rolling(window=50).mean()
3 plt.figure(figsize=(14, 7))
4 plt.plot(df['Close'].iloc[:, 0], label='Price')
5 plt.plot(df['MA20'], label='20-Day MA')
6 plt.plot(df['MA50'], label='50-Day MA')
7 plt.title('Bitcoin Price with Moving Averages')
8 plt.legend()
9 plt.grid(True)
10 plt.show()
```

Bitcoin Price with Moving Averages



Relative Strength Index (RSI)

```
1 delta = df['Close'].iloc[:, 0].diff()
2 gain = (delta.where(delta > 0, 0)).rolling(window=14).mean()
3 loss = (-delta.where(delta < 0, 0)).rolling(window=14).mean()
4 rs = gain / loss
5 df['RSI'] = 100 - (100 / (1 + rs))
6 plt.figure(figsize=(14, 4))
7 plt.plot(df.index, df['RSI'], color='purple')
8 plt.axhline(70, linestyle='--', alpha=0.5, color='red')
9 plt.axhline(30, linestyle='--', alpha=0.5, color='green')
10 plt.title('RSI Indicator')
11 plt.ylim(0, 100)
12 plt.grid(True)
13 plt.show()
```



```
1 !pip install -q prophet
```

```
1 from prophet import Prophet
2 train_df = df['Close'].iloc[:, 0].reset_index()
3 train_df.columns = ['ds', 'y']
4 train_df['ds'] = train_df['ds'].dt.tz_localize(None)
5 display(train_df.tail())
```

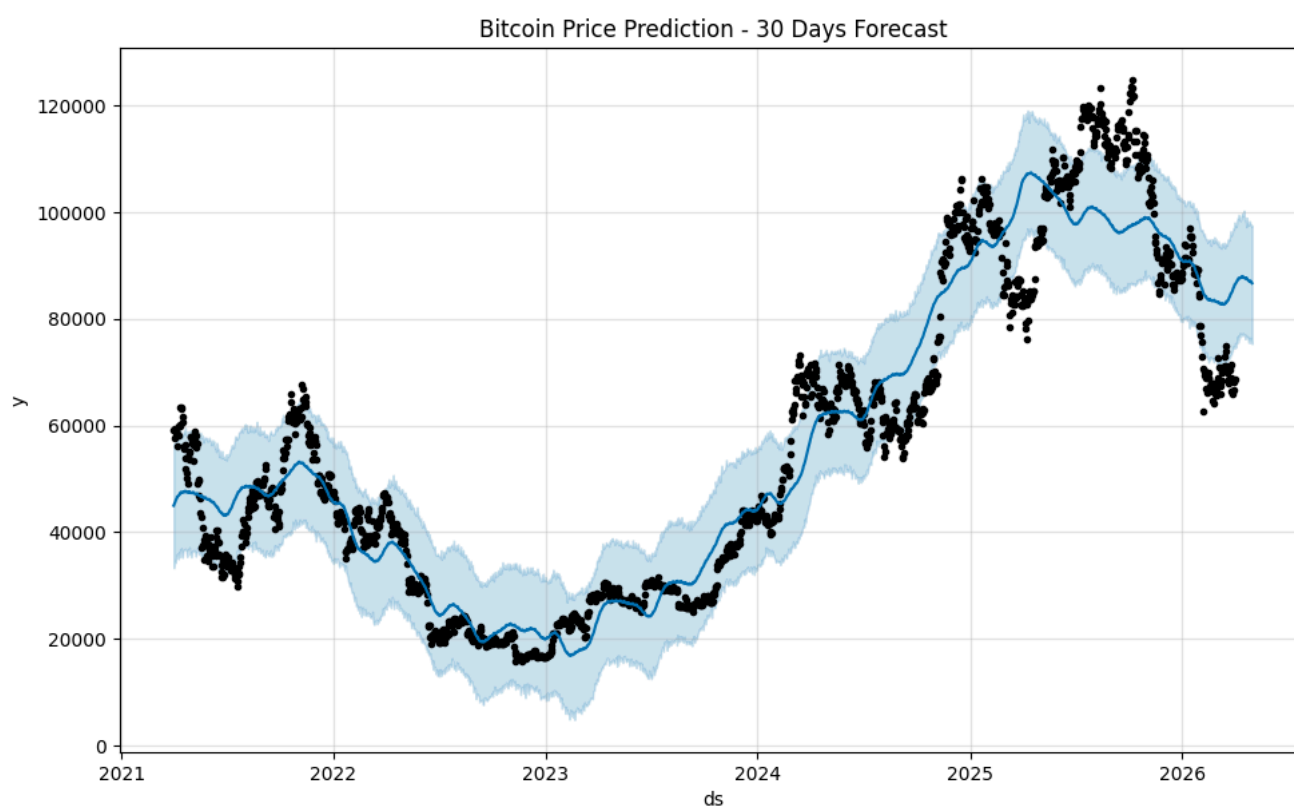
	ds	y
1822	2026-03-28	66319.695312
1823	2026-03-29	65954.921875
1824	2026-03-30	66691.445312
1825	2026-03-31	68233.312500
1826	2026-04-01	68575.265625

```
1 model = Prophet(daily_seasonality=True, yearly_seasonality=True)
2 model.fit(train_df)
```

```
<prophet.forecaster.Prophet at 0x78f56a1fab40>
```

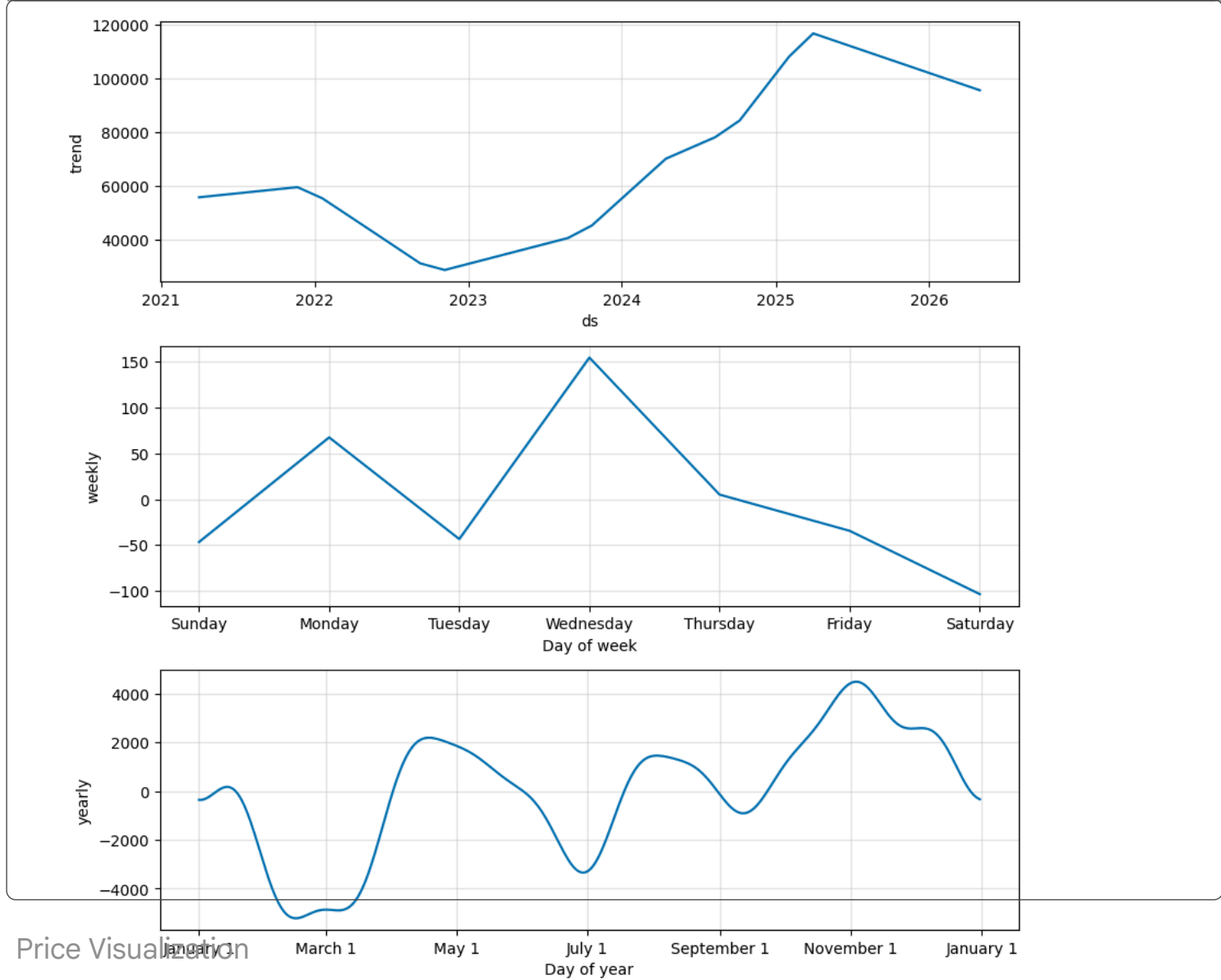
Price Forecasting

```
1 future = model.make_future_dataframe(periods=30)
2 forecast = model.predict(future)
3 fig1 = model.plot(forecast)
4 plt.title('Bitcoin Price Prediction - 30 Days Forecast')
5 plt.show()
```



Forecast Components

```
1 fig2 = model.plot_components(forecast)
2 plt.show()
```



Price Visualization

```
1 import matplotlib.pyplot as plt
2 plt.figure(figsize=(14, 7))
3 plt.plot(df['Close'])
4 plt.title('Bitcoin Price over Time')
5 plt.xlabel('Date')
6 plt.ylabel('Price (USD)')
7 plt.grid(True)
8 plt.show()
```

